

hw3

April 30, 2026

```
[1]: import project3 as p1
import utils
import numpy as np

def plot_toy_results(algo_name, thetas, toy_features, toy_labels):
    print('theta for', algo_name, 'is', ', '.join(map(str, list(thetas[0]))))
    print('theta_0 for', algo_name, 'is', str(thetas[1]))
    utils.plot_toy_data(algo_name, toy_features, toy_labels, thetas)

def main(file_name):
    toy_features, toy_labels = toy_data = utils.load_toy_data(file_name)

    T = 10
    L = 0.2

    thetas_perceptron = p1.perceptron(toy_features, toy_labels, T)
    thetas_avg_perceptron = p1.average_perceptron(toy_features, toy_labels, T)
    thetas_pegasos = p1.pegasos(toy_features, toy_labels, T, L)

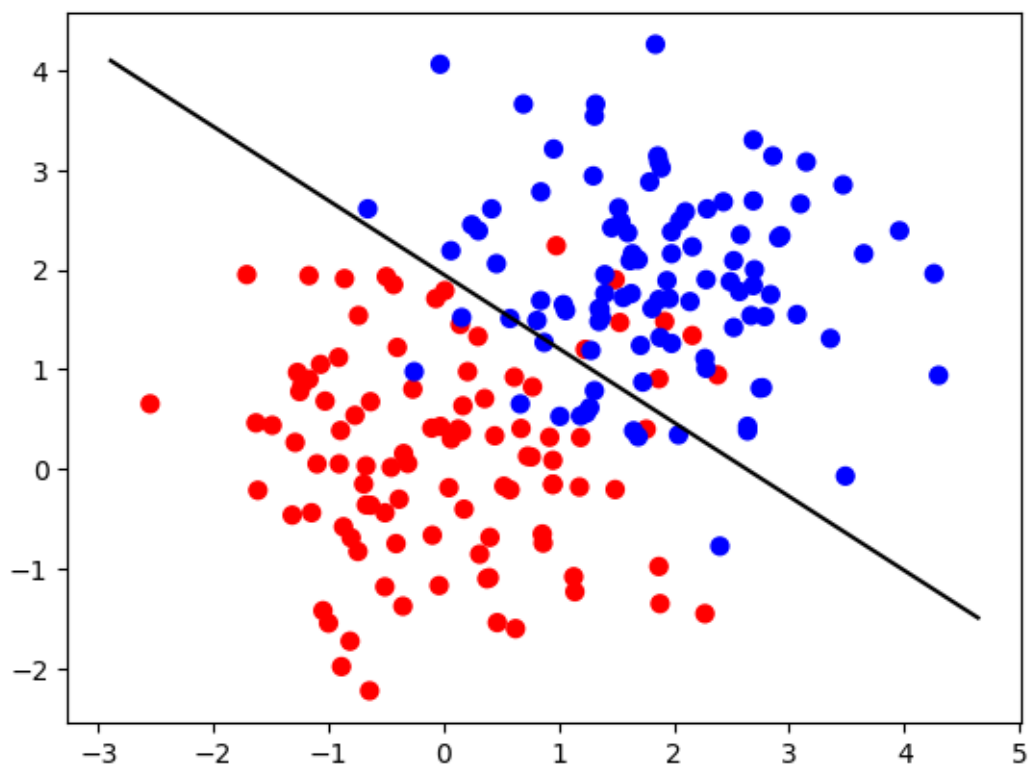
    # Visualization part
    # You can check the figures and test the codes for your own purpose
    # This will not be graded
    plot_toy_results('Perceptron', thetas_perceptron, toy_features, toy_labels)
    plot_toy_results('Average Perceptron', thetas_avg_perceptron, toy_features,
    ↪toy_labels)
    plot_toy_results('Pegasos', thetas_pegasos, toy_features, toy_labels)

if __name__ == '__main__':
    main("toy_data.tsv")
```

theta for Perceptron is 2.280399999999993, 3.074999999999975

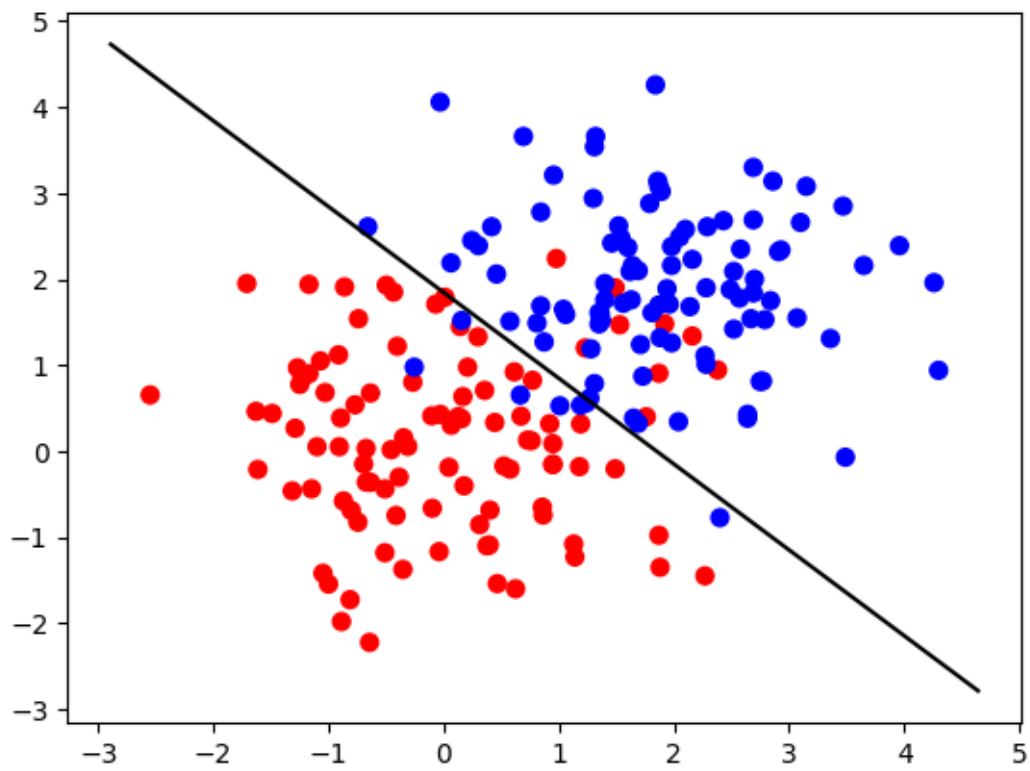
theta_0 for Perceptron is -6.0

Classified Toy Data (Perceptron)



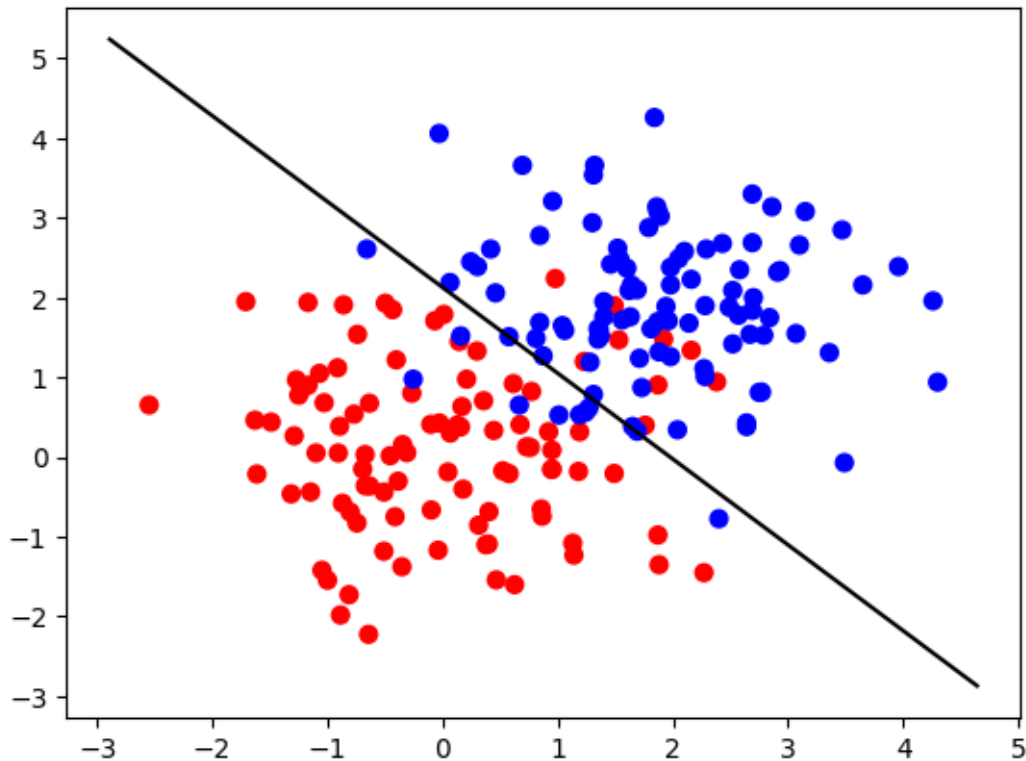
theta for Average Perceptron is 3.0792973499999876, 3.0865134999999997
theta_0 for Average Perceptron is -5.6965

Classified Toy Data (Average Perceptron)



theta for Pegasos is 0.637685075696591, 0.5924832274128278
theta_0 for Pegasos is -1.2607734140059579

Classified Toy Data (Pegasos)



```
[5]: def main(file_name):
    toy_features, toy_labels = toy_data = utils.load_toy_data(file_name)

    T = int(1e5)
    L = 0.2

    thetas_perceptron = p1.perceptron(toy_features, toy_labels, T)
    thetas_avg_perceptron = p1.average_perceptron(toy_features, toy_labels, T)
    thetas_pegasos = p1.pegasos(toy_features, toy_labels, T, L)

    # Visualization part
    # You can check the figures and test the codes for your own purpose
    # This will not be graded
    plot_toy_results('Perceptron', thetas_perceptron, toy_features, toy_labels)
    plot_toy_results('Average Perceptron', thetas_avg_perceptron, toy_features,
    ↪toy_labels)
    plot_toy_results('Pegasos', thetas_pegasos, toy_features, toy_labels)

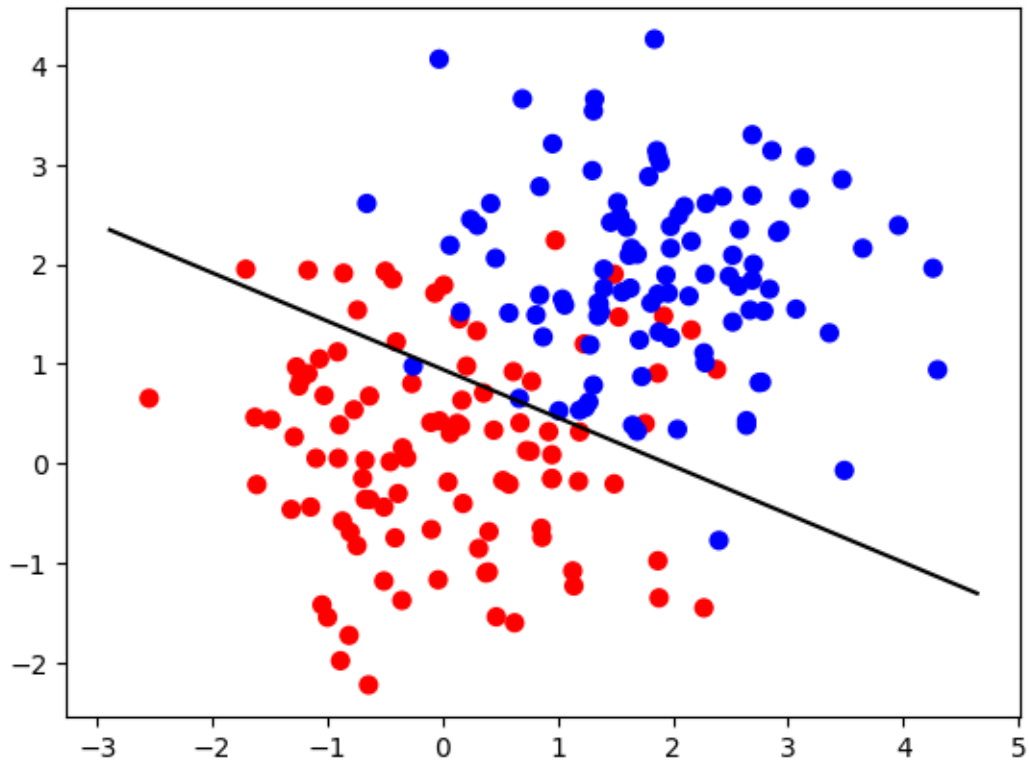
    if __name__ == '__main__':
```

```
main("toy_data.tsv")
```

theta for Perceptron is 2.051699999957741, 4.236199999991653

theta_0 for Perceptron is -4.0

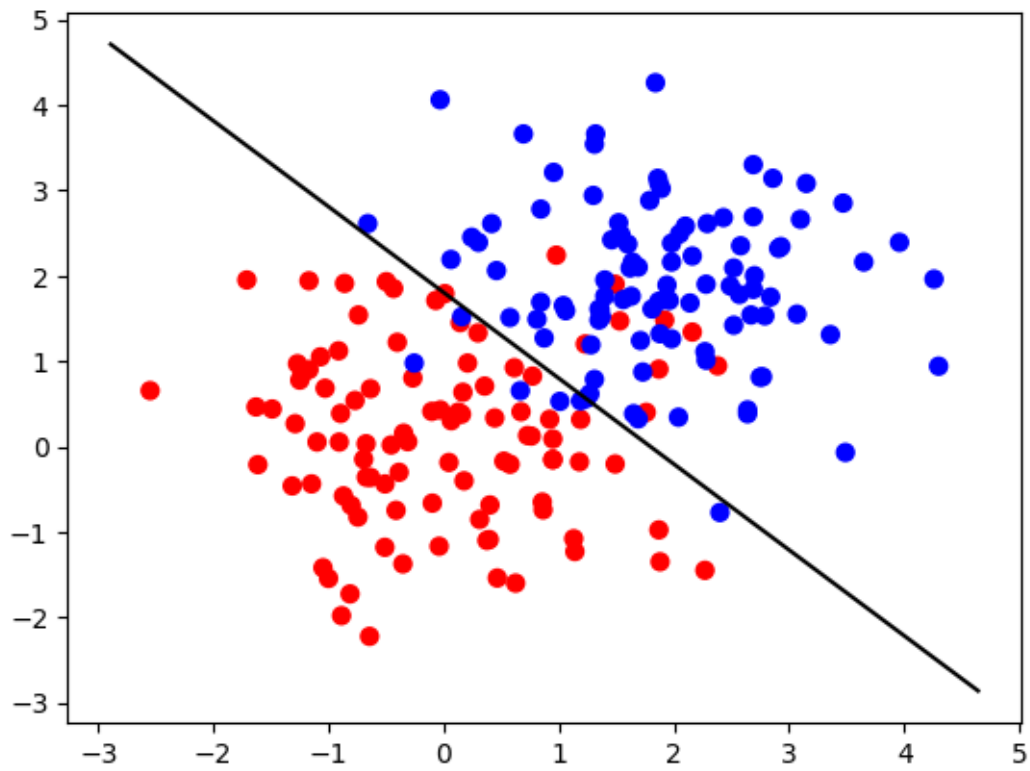
Classified Toy Data (Perceptron)



theta for Average Perceptron is 3.1069823466544415, 3.0925295192433944

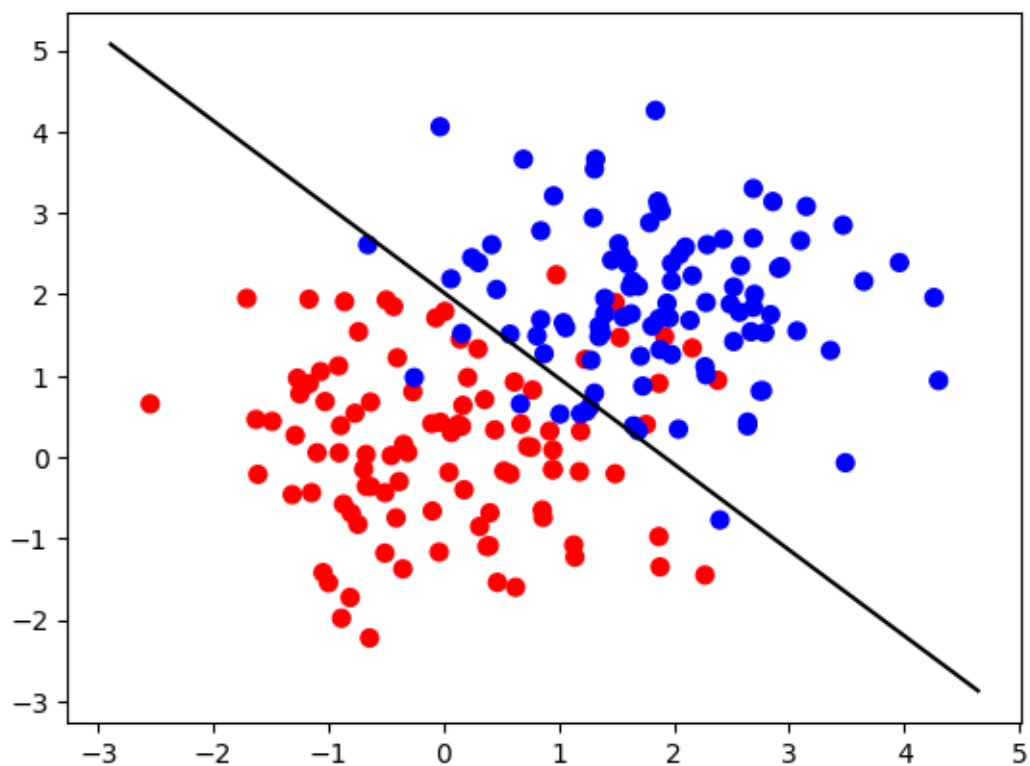
theta_0 for Average Perceptron is -5.5771573

Classified Toy Data (Average Perceptron)



theta for Pegasos is 0.6407449110324509, 0.6076639305855491
theta_0 for Pegasos is -1.229099043528132

Classified Toy Data (Pegasos)



Average Perceptron and Pegasos converge on this dataset. Standard Perceptron does not clearly converge, since its parameters continue to fluctuate as T increases.